

LINUX COMMANDS TASK -2

1. Create user with name Techie and provide sudo access to user.

Used useradd command to create a new user with name Techie and provided sudo access to the user by editing sudoers file and providing ALL access in the group user section for this user.

```
vboxuser@Ubuntu:~/Desktop$ useradd Techie
useradd: Permission denied.
useradd: cannot lock /etc/passwd; try again later.
vboxuser@Ubuntu:~/Desktop$ sudo useradd Techie
vboxuser@Ubuntu:~/Desktop$ sudo passwd Techie
New password:
BAD PASSWORD: The password fails the dictionary check - it is too simplistic/systematic
Retype new password:
passwd: password updated successfully
vboxuser@Ubuntu:~/Desktop$ sudo su -
root@Ubuntu:~# vi /etc/sudoers
root@Ubuntu:~# su Techie
$ mkdir Linux1
mkdir: Permission denied
$ sudo mkdir Linux1
[sudo: authenticate] Password:
$
```

2. Navigate to the home directory : Used cd /home command.
3. Create a new directory : Used mkdir to create new directory.
4. List the contents of a directory : Used ll command to list the contents.

```
vboxuser@Ubuntu:/$ cd /home
vboxuser@Ubuntu:/home$ sudo mkdir Techie1
[sudo: authenticate] Password:
vboxuser@Ubuntu:/home$ ls
Linux Techie1 vboxuser
vboxuser@Ubuntu:/home$ ll
total 20
drwxr-xr-x  5 root    root    4096 Jun 25 09:47 ./
drwxr-xr-x 20 root    root    4096 Jun 25 09:41 ../
drwxr-xr-x  2 root    root    4096 Jun 25 09:39 Linux/
drwxr-xr-x  2 root    root    4096 Jun 25 09:47 Techie1/
drwxr-x--- 15 vboxuser vboxuser 4096 Jun 25 09:46 vboxuser/
vboxuser@Ubuntu:/home$
```

5. Change the current directory : Used `cd ..` command to change the current directory.
6. Create a new empty file : Used `touch` command to create a new file with 0 bytes.
7. View the contents of a file : Used `cat` command to view the contents of the file and `wc` command to count the characters in the file.

```
vboxuser@Ubuntu:/home/Techie1$ sudo touch Linux
vboxuser@Ubuntu:/home/Techie1$ sudo vi Linux
vboxuser@Ubuntu:/home/Techie1$ cat Linux
abcdefghijklmnopqrstuvwxyz

vboxuser@Ubuntu:/home/Techie1$ wc Linux
 2  1 28 Linux
vboxuser@Ubuntu:/home/Techie1$
```

8. Copy a file to another location : Used `cp` command to copy file to another location.
9. Move a file to another location : Used `mv` command to move file to other location.

```
vboxuser@Ubuntu:/home/Techie1$ cp Linux /home
cp: cannot create regular file '/home/Linux': Permission denied
vboxuser@Ubuntu:/home/Techie1$ sudo cp Linux /home
vboxuser@Ubuntu:/home/Techie1$ sudo mv Linux Techie1
```

10. Rename a file : Used mv command to rename a file.

11. Delete a file : Used rm command to delete a file.

```
vboxuser@Ubuntu:/home/Techie1$ cd ..
vboxuser@Ubuntu:/home$ mv Linux Linux1
mv: cannot move 'Linux' to 'Linux1': Permission denied
vboxuser@Ubuntu:/home$ sudo mv Linux Linux1
vboxuser@Ubuntu:/home$ rm Linux1
rm: remove write-protected regular file 'Linux1'? y
rm: cannot remove 'Linux1': Permission denied
vboxuser@Ubuntu:/home$ sudo rm Linux1
```

12. Grant or revoke permissions on a file or directory : Used chmod 777 command to grant full permissions to all users for the Linux file.

```
vboxuser@Ubuntu:/home/Techie1$ ls
Linux  Techie1
vboxuser@Ubuntu:/home/Techie1$ ll
total 16
drwxr-xr-x 2 root root 4096 Jun 25 10:15 ./
drwxr-xr-x 4 root root 4096 Jun 25 10:12 ../
-rw-r--r-- 1 root root  27 Jun 25 10:15 Linux
-rw-r--r-- 1 root root  28 Jun 25 10:01 Techie1
vboxuser@Ubuntu:/home/Techie1$ sudo chmod 777
chmod: missing operand
Try 'chmod --help' for more information.
vboxuser@Ubuntu:/home/Techie1$ sudo chmod 777 Linux
vboxuser@Ubuntu:/home/Techie1$ ll
total 16
drwxr-xr-x 2 root root 4096 Jun 25 10:15 ./
drwxr-xr-x 4 root root 4096 Jun 25 10:12 ../
-rwxrwxrwx 1 root root  27 Jun 25 10:15 Linux*
-rw-r--r-- 1 root root  28 Jun 25 10:01 Techie1
```

13. View the current date and time : Used date command.

14. Check the system uptime : Used uptime -p and uptime -s for system uptime details.

15. View the running processes : Used ps -ef to view all the running processes.

```
vboxuser@Ubuntu:/home/Techie1$ date
Thu Jun 25 10:20:15 AM UTC 2026
vboxuser@Ubuntu:/home/Techie1$ uptime -p
up 1 hour, 20 minutes
vboxuser@Ubuntu:/home/Techie1$ uptime -s
2026-06-25 09:03:29
vboxuser@Ubuntu:/home/Techie1$ ps -ef
```

UID	PID	PPID	C	STIME	TTY	TIME	CMD
root	1	0	0	09:03	?	00:00:07	/usr/lib/systemd/systemd --switched-root --system --deserialize=51
root	2	0	0	09:03	?	00:00:00	[kthreadd]
root	3	2	0	09:03	?	00:00:00	[pool_workqueue_release]
root	4	2	0	09:03	?	00:00:00	[kworker/R-rcu_gp]
root	5	2	0	09:03	?	00:00:00	[kworker/R-sync_wq]
root	6	2	0	09:03	?	00:00:00	[kworker/R-kvfree_rcu_reclaim]
root	7	2	0	09:03	?	00:00:00	[kworker/R-slub_flushwq]
root	8	2	0	09:03	?	00:00:00	[kworker/R-netns]
root	10	2	0	09:03	?	00:00:00	[kworker/0:0H-kblockd]
root	13	2	0	09:03	?	00:00:00	[kworker/R-mm_percpu_wq]
root	14	2	0	09:03	?	00:00:00	[ksoftirqd/0]
root	15	2	0	09:03	?	00:00:01	[rcu_preempt]
root	16	2	0	09:03	?	00:00:00	[rcu_exp_par_gp_kthread_worker/0]
root	17	2	0	09:03	?	00:00:00	[rcu_exp_gp_kthread_worker]

16. Kill a running process : Used kill -9 PID command to kill that process.

```
vboxuser 5008 5002 0 09:32 pts/6 00:00:00 bash
root 5039 5008 0 09:35 pts/6 00:00:00 su Techie
Techie 5043 5039 0 09:36 pts/6 00:00:00 sh
root 5050 5043 0 09:37 pts/6 00:00:00 sudo su root
root 5052 5050 0 09:37 pts/7 00:00:00 sudo su root
root 5053 5052 0 09:37 pts/7 00:00:00 su root
root 5055 5053 0 09:37 pts/7 00:00:00 bash
root 5080 2 0 09:40 ? 00:00:03 [kworker/0:0-events]
root 5172 2 0 09:45 ? 00:00:02 [kworker/u8:2-events_power_efficient]
vboxuser 5403 3712 1 09:55 ? 00:00:19 ptyxis --new-window --working-directory /home/vboxuser/Desktop
vboxuser 5417 5403 0 09:55 ? 00:00:01 /usr/libexec/ptyxis-agent --socket-fd=3 --rlimit-nofile=1024
vboxuser 5448 5417 0 09:55 pts/4 00:00:00 /usr/bin/bash
vboxuser 5449 5417 0 09:55 pts/8 00:00:00 /usr/bin/bash
root 5473 2 0 09:55 ? 00:00:00 [kworker/1:1-events]
root 5499 2 0 09:58 ? 00:00:02 [kworker/u8:3-events_power_efficient]
root 5581 2 0 10:05 ? 00:00:00 [kworker/0:2-rcu_gp]
root 5624 2 0 10:07 ? 00:00:01 [kworker/u8:4-events_unbound]
root 5722 2 0 10:12 ? 00:00:00 [kworker/1:0-cgroup_free]
root 5838 2 0 10:20 ? 00:00:00 [kworker/0:1-cgroup_release]
root 5840 2 0 10:20 ? 00:00:00 [kworker/0:3-cgroup_free]
root 5852 2 0 10:20 ? 00:00:00 [kworker/1:2]
vboxuser 5864 5448 99 10:25 pts/4 00:00:00 ps -ef
vboxuser@Ubuntu:/home/Techie1$ kill -9 5722
bash: kill: (5722) - No such process
vboxuser@Ubuntu:/home/Techie1$ kill -9 5043
bash: kill: (5043) - Operation not permitted
vboxuser@Ubuntu:/home/Techie1$ sudo kill -9 5043
```

17. Install a package using the package manager (e.g., apt or yum).

Used apt install git command to install the git package.


```
vboxuser@ubuntu:/home/Techie1$ sudo apt install git
The following packages were automatically installed and are no longer required:
  linux-headers-7.0.0-14          linux-image-unsigned-7.0.0-14-generic  linux-modules-7.0.0-14-generic  linux-tools-7.0.0-14-generic
  linux-headers-7.0.0-14-generic  linux-main-modules-zfs-7.0.0-14-generic  linux-tools-7.0.0-14
Use 'sudo apt autoremove' to remove them.

Installing:
  git

Installing dependencies:
  git-man  liberror-perl

Suggested packages:
  git-doc  git-email  git-gui  gitk  gitweb  git-cvs  git-svn

Summary:
  Upgrading: 0, Installing: 3, Removing: 0, Not Upgrading: 30
  Download size: 5,453 kB
  Space needed: 28.4 MB / 17.6 GB available

Continue? [Y/n] y
Get:1 http://in.archive.ubuntu.com/ubuntu resolute/main amd64 liberror-perl all 0.17030-1 [23.5 kB]
Get:2 http://in.archive.ubuntu.com/ubuntu resolute/main amd64 git-man all 1:2.53.0-1ubuntu1 [1,202 kB]
Get:3 http://in.archive.ubuntu.com/ubuntu resolute/main amd64 git amd64 1:2.53.0-1ubuntu1 [4,227 kB]
Fetched 5,453 kB in 4s (1,515 kB/s)
Selecting previously unselected package liberror-perl.
(Reading database... 191751 files and directories currently installed.)
Preparing to unpack ./liberror-perl_0.17030-1_all.deb...
Unpacking liberror-perl (0.17030-1)...
Selecting previously unselected package git-man.
Preparing to unpack ./git-man_1%3a2.53.0-1ubuntu1_all.deb...
Unpacking git-man (1:2.53.0-1ubuntu1)...
Selecting previously unselected package git.
Preparing to unpack ./git_1%3a2.53.0-1ubuntu1_amd64.deb...
Unpacking git (1:2.53.0-1ubuntu1)...
Setting up liberror-perl (0.17030-1)...
Setting up git-man (1:2.53.0-1ubuntu1)...
Setting up git (1:2.53.0-1ubuntu1)...
```

18. Update the system packages: Used apt update and apt upgrade -y command to see the updates available and also install the updates for the available system packages.

```
vboxuser@ubuntu:/home/Techie1$ sudo apt update
Hit:1 http://in.archive.ubuntu.com/ubuntu resolute InRelease
Get:2 http://security.ubuntu.com/ubuntu resolute-security InRelease [137 kB]
Get:3 http://in.archive.ubuntu.com/ubuntu resolute-updates InRelease [137 kB]
Get:4 http://security.ubuntu.com/ubuntu resolute-security/main amd64 Packages [232 kB]
Hit:5 http://in.archive.ubuntu.com/ubuntu resolute-backports InRelease
Get:6 http://security.ubuntu.com/ubuntu resolute-security/main Translation-en [62.2 kB]
Get:7 http://security.ubuntu.com/ubuntu resolute-security/main amd64 Components [31.2 kB]
Get:8 http://security.ubuntu.com/ubuntu resolute-security/main amd64 c-n-f Metadata [3,920 B]
Get:9 http://security.ubuntu.com/ubuntu resolute-security/restricted amd64 Packages [208 kB]
Get:10 http://security.ubuntu.com/ubuntu resolute-security/restricted Translation-en [36.1 kB]
Get:11 http://security.ubuntu.com/ubuntu resolute-security/universe amd64 Packages [126 kB]
Get:12 http://security.ubuntu.com/ubuntu resolute-security/universe Translation-en [39.1 kB]
Get:13 http://security.ubuntu.com/ubuntu resolute-security/universe amd64 Components [43.1 kB]
Get:14 http://security.ubuntu.com/ubuntu resolute-security/universe amd64 c-n-f Metadata [3,288 B]
Get:15 http://in.archive.ubuntu.com/ubuntu resolute-updates/main amd64 Packages [266 kB]
Get:16 http://in.archive.ubuntu.com/ubuntu resolute-updates/main Translation-en [71.8 kB]
Get:17 http://in.archive.ubuntu.com/ubuntu resolute-updates/main amd64 Components [49.4 kB]
Get:18 http://in.archive.ubuntu.com/ubuntu resolute-updates/main amd64 c-n-f Metadata [5,140 B]
Get:19 http://in.archive.ubuntu.com/ubuntu resolute-updates/universe amd64 Packages [153 kB]
Get:20 http://in.archive.ubuntu.com/ubuntu resolute-updates/universe Translation-en [48.3 kB]
Get:21 http://in.archive.ubuntu.com/ubuntu resolute-updates/universe amd64 Components [63.0 kB]
Get:22 http://in.archive.ubuntu.com/ubuntu resolute-updates/universe Icons (64x64) [43.5 kB]
Get:23 http://in.archive.ubuntu.com/ubuntu resolute-updates/universe amd64 c-n-f Metadata [3,956 B]
Fetched 1,764 kB in 2s (898 kB/s)
40 packages can be upgraded. Run 'apt list --upgradable' to see them.
vboxuser@ubuntu:/home/Techie1$ sudo apt upgrade -y
The following packages were automatically installed and are no longer required:
  linux-headers-7.0.0-14          linux-image-unsigned-7.0.0-14-generic  linux-modules-7.0.0-14-generic  linux-tools-7.0.0-14-generic
  linux-headers-7.0.0-14-generic  linux-main-modules-zfs-7.0.0-14-generic  linux-tools-7.0.0-14
Use 'sudo apt autoremove' to remove them.

Upgrading:
  accountsservice          gnome-control-center          libaccountsservice0          libmutter-18-0          mutter-common-bin          snapd
  alsa-ucm-conf             gnome-control-center-data     libgtk-4-1                   libnfs14                perl                      ubuntu-release-upgrader-core
  base-files                gnome-control-center-faces    libgtk-4-bin                 libperl5.40             perl-base                 ubuntu-release-upgrader-gtk
  glib1.2-accountsservice-1.0  gnome-session-bin            libgtk-4-common              libxml2-10              perl-modules-5.40         ubuntu-session
  glib1.2-gtk-4-0           gnome-session-common          libical4-common              locales                  python3-distupgrade       update-notifier
```

19. Create a symbolic link : Used ln -s command to create a link.

```
lin: Permission denied
vboxuser@Ubuntu:/home/Techie1$ sudo ln -s /home
vboxuser@Ubuntu:/home/Techie1$ ls
Linux Techie1 home
```

20. Search for files using the find command : Used find / -name Linux command to search for Linux name file.

```
vboxuser@Ubuntu:/home$ sudo find / -name Linux
/home/Techie1/Linux
find: '/run/user/1000/gvfs': Permission denied
find: '/run/user/1000/doc': Permission denied
vboxuser@Ubuntu:/home$
```

21. Compress and decompress files using tar.

Tar cvf used to create verbose file and bundle it into a single file.

Tar xvf used to extract verbose file and see the contents or split the bundled file.

```
vboxuser@Ubuntu:~/Desktop$ ls
a b c d e f g h
vboxuser@Ubuntu:~/Desktop$ tar -czvf backup.tar.gz a
a
vboxuser@Ubuntu:~/Desktop$ ls
a b backup.tar.gz c d e f g h
vboxuser@Ubuntu:~/Desktop$ tar -xzvf backup.tar.gz
a
vboxuser@Ubuntu:~/Desktop$ ls
a b backup.tar.gz c d e f g h
vboxuser@Ubuntu:~/Desktop$
```

22. Monitor system resources with top or htop.

Top command used to show and monitor the system resources.

```
top - 04:10:07 up 12 min, 1 user, load average: 0.32, 0.33, 0.32
Tasks: 193 total, 1 running, 191 sleeping, 0 stopped, 1 zombie
%Cpu(s): 0.1 us, 0.4 sy, 0.0 ni, 99.3 id, 0.0 wa, 0.0 hi, 0.2 si, 0.0 st
MiB Mem : 7424.8 total, 4852.7 free, 1229.1 used, 1613.0 buff/cache
MiB Swap: 0.0 total, 0.0 free, 0.0 used. 6195.7 avail Mem
```

PID	USER	PR	NI	VIRT	RES	SHR	S	%CPU	%MEM	TIME+	COMMAND
4008	vboxuser	20	0	1781996	361876	146544	S	1.2	4.8	0:08.71	ptxis
3031	vboxuser	20	0	3884716	345352	168096	S	0.7	4.5	0:16.78	gnome-shell
35	root	20	0	0	0	0	I	0.2	0.0	0:01.06	kworker/u8:1-events_power_efficient
37	root	20	0	0	0	0	I	0.2	0.0	0:01.40	kworker/u8:2-flush-8:0
70	root	20	0	0	0	0	I	0.2	0.0	0:01.17	kworker/0:2-events
1182	_chrony	20	0	20560	6256	5364	S	0.2	0.1	0:00.41	chronyd
1	root	20	0	25548	16624	11640	S	0.0	0.2	0:06.63	systemd
2	root	20	0	0	0	0	S	0.0	0.0	0:00.02	kthread
3	root	20	0	0	0	0	S	0.0	0.0	0:00.00	pool_workqueue_release
4	root	0	-20	0	0	0	I	0.0	0.0	0:00.00	kworker/R-rcu_gp
5	root	0	-20	0	0	0	I	0.0	0.0	0:00.00	kworker/R-sync_wq
6	root	0	-20	0	0	0	I	0.0	0.0	0:00.00	kworker/R-kvfree_rcu_reclaim
7	root	0	-20	0	0	0	I	0.0	0.0	0:00.00	kworker/R-slub_flushwq
8	root	0	-20	0	0	0	I	0.0	0.0	0:00.00	kworker/R-netns
9	root	20	0	0	0	0	I	0.0	0.0	0:00.62	kworker/0:0-cgwb_release
10	root	0	-20	0	0	0	I	0.0	0.0	0:00.42	kworker/0:0H-kblockd
13	root	0	-20	0	0	0	I	0.0	0.0	0:00.00	kworker/R-mm_percpu_wq
14	root	20	0	0	0	0	S	0.0	0.0	0:00.22	ksoftirqd/0
15	root	20	0	0	0	0	I	0.0	0.0	0:00.54	rcu_preempt
16	root	20	0	0	0	0	S	0.0	0.0	0:00.00	rcu_exp_par_gp_kthread_worker/0
17	root	20	0	0	0	0	S	0.0	0.0	0:00.10	rcu_exp_gp_kthread_worker
18	root	rt	0	0	0	0	S	0.0	0.0	0:00.01	migration/0
19	root	20	0	0	0	0	S	0.0	0.0	0:00.00	kprobe-optimizer
20	root	-51	0	0	0	0	S	0.0	0.0	0:00.00	idle_inject/0
21	root	20	0	0	0	0	S	0.0	0.0	0:00.00	cpuhp/0
22	root	20	0	0	0	0	S	0.0	0.0	0:00.00	cpuhp/1
23	root	-51	0	0	0	0	S	0.0	0.0	0:00.00	idle_inject/1
24	root	rt	0	0	0	0	S	0.0	0.0	0:00.02	migration/1
25	root	20	0	0	0	0	S	0.0	0.0	0:00.25	ksoftirqd/1
26	root	20	0	0	0	0	I	0.0	0.0	0:00.27	kworker/1:0-events

23. Create and manage user groups.

Used sudo groupadd to create a user group named devops

Used sudo usermod -aG devops Techie to add Techie user to the user group devops.

```
vboxuser@Ubuntu:~/Desktop$ sudo groupadd devops
vboxuser@Ubuntu:~/Desktop$ sudo usermod -aG devops Techie
vboxuser@Ubuntu:~/Desktop$
```

24. Set up SSH password less authentication.

Configured ssh key-based authentication so that users can log into remote server without entering a password.

EX: Amazon Linux

```
salma@SALMANBABSAIL MINGW64 ~/Downloads
$ ssh -i Linux_test1.pem ec2-user@3.95.58.144
The authenticity of host '3.95.58.144 (3.95.58.144)' can't be established.
ED25519 key fingerprint is SHA256:ryfkCwB721j2Vj1lrKsBw4rrfydu7zTZFm1GTi7iQCM.
This key is not known by any other names
Are you sure you want to continue connecting (yes/no/[fingerprint])? y
Please type 'yes', 'no' or the fingerprint: yes
Warning: Permanently added '3.95.58.144' (ED25519) to the list of known hosts.

#_
~\##### Amazon Linux 2023
~~\#####
~~\###|
~~\#/
~~V~'-'> https://aws.amazon.com/linux/amazon-linux-2023
~~~
~~~
~~~
~/m/'

Last login: Mon Jun 29 09:12:16 2026 from 18.206.107.27
[ec2-user@ip-172-31-23-85 ~]$ |
```

25. Monitor log files using tail or grep.

Used tail and grep command to monitor and analyze linux log files in real time for system administration and troubleshooting.

```
vboxuser@Ubuntu:~/Desktop$ tail -f /var/log/syslog
2026-06-29T04:05:43.797596+00:00 Ubuntu kernel: audit: type=1400 audit(1782705943.796:265): apparmor="AUDIT" operation="usersns_create" class="namespac
e" info="usersns create - transitioning profile" profile="unconfined" pid=4285 comm="(ubuntu-insigh)" requested="usersns_create" target="unprivileged_us
erns" execpath="/usr/lib/systemd/systemd-executor"
2026-06-29T04:05:43.806352+00:00 Ubuntu kernel: audit: type=1400 audit(1782705943.796:266): apparmor="DENIED" operation="capable" class="cap" profile=
"unprivileged_usersns" pid=4285 comm="(ubuntu-insigh)" capability=21 capname="sys_admin"
2026-06-29T04:05:43.836534+00:00 Ubuntu systemd[2736]: Finished ubuntu-insights-upload.service - "Upload collected and matured platform reports using
Ubuntu Insights while respecting consent".
2026-06-29T04:10:11.296639+00:00 Ubuntu systemd[1]: apt-daily-upgrade.service - Daily apt upgrade and clean activities skipped, unmet condition check
ConditionACPower=true
2026-06-29T04:10:11.311793+00:00 Ubuntu systemd[1]: Starting sysstat-collect.service - system activity accounting tool...
2026-06-29T04:10:11.352589+00:00 Ubuntu systemd[1]: sysstat-collect.service: Deactivated successfully.
2026-06-29T04:10:11.352779+00:00 Ubuntu systemd[1]: Finished sysstat-collect.service - system activity accounting tool.
2026-06-29T04:12:43.775260+00:00 Ubuntu systemd[1]: Starting systemd-tmpfiles-clean.service - Cleanup of Temporary Directories...
2026-06-29T04:12:43.976567+00:00 Ubuntu systemd[1]: systemd-tmpfiles-clean.service: Deactivated successfully.
2026-06-29T04:12:43.976653+00:00 Ubuntu systemd[1]: Finished systemd-tmpfiles-clean.service - Cleanup of Temporary Directories.

vboxuser@Ubuntu:~/Desktop$ grep ERROR /var/log/syslog
2026-06-22T07:14:36.127059+00:00 Ubuntu kernel: vmwgfx 0000:00:02.0: [drm] *ERROR* vmwgfx seems to be running on an unsupported hypervisor.
2026-06-22T07:14:36.127059+00:00 Ubuntu kernel: vmwgfx 0000:00:02.0: [drm] *ERROR* This configuration is likely broken.
2026-06-22T07:14:36.127060+00:00 Ubuntu kernel: vmwgfx 0000:00:02.0: [drm] *ERROR* Please switch to a supported graphics device to avoid problems.
grep: /var/log/syslog: binary file matches
vboxuser@Ubuntu:~/Desktop$
```

26. Set up a web server (e.g., Apache or Nginx).

Installed, configured and ran Apache on Ubuntu

```
vboxuser@Ubuntu:~/Desktop$ sudo apt update && sudo apt install apache2 -y && sudo systemctl start apache2 && sudo systemctl enable apache2
Hit:1 http://in.archive.ubuntu.com/ubuntu resolute InRelease
Ign:2 https://pkgs.jenkins.io/debian-stable binary/ InRelease
Hit:3 https://pkgs.jenkins.io/debian-stable binary/ Release
Get:5 http://security.ubuntu.com/ubuntu resolute-security InRelease [137 kB]
Get:6 http://in.archive.ubuntu.com/ubuntu resolute-updates InRelease [137 kB]
Get:7 http://security.ubuntu.com/ubuntu resolute-security/main amd64 Components [31.2 kB]
Get:8 http://security.ubuntu.com/ubuntu resolute-security/universe amd64 Components [43.2 kB]
Hit:9 http://in.archive.ubuntu.com/ubuntu resolute-backports InRelease
Get:10 http://in.archive.ubuntu.com/ubuntu resolute-updates/main amd64 Packages [282 kB]
Get:11 http://in.archive.ubuntu.com/ubuntu resolute-updates/main amd64 Components [49.4 kB]
Get:12 http://in.archive.ubuntu.com/ubuntu resolute-updates/universe amd64 Packages [159 kB]
Get:13 http://in.archive.ubuntu.com/ubuntu resolute-updates/universe amd64 Components [62.9 kB]
Fetched 902 kB in 2s (446 kB/s)
6 packages can be upgraded. Run 'apt list --upgradable' to see them.
Installing:
  apache2
Installing dependencies:
  apache2-bin  apache2-data  apache2-utils  libapr1t64  libaprutil1-dbd-sqlite3  libaprutil1-ldap  libaprutil1t64
Suggested packages:
  apache2-doc  apache2-suexec-pristine  | apache2-suexec-custom
Summary:
  Upgrading: 0, Installing: 0, Removing: 0, Not Upgrading: 0
  Download size: 1,935 kB
  Space needed: 7,565 kB / 16.8 GB available
```



```
vboxuser@Ubuntu:~/Desktop$ sudo systemctl status apache2
● apache2.service - The Apache HTTP Server
   Loaded: loaded (/usr/lib/systemd/system/apache2.service; enabled; preset: enabled)
   Active: active (running) since Mon 2026-06-29 04:16:08 UTC; 59s ago
     Invocation: c67edfcf973942fa9eb81f525eb2cb85
       Docs: https://httpd.apache.org/docs/2.4/
    Main PID: 5385 (apache2)
      Status: "Total requests: 0; Idle/Busy workers 100/0;Requests/sec: 0; Bytes served/sec: 0 B/sec"
        Tasks: 55 (limit: 6371)
       Memory: 5.4M (peak: 5.9M)
         CPU: 156ms
      CGroup: /system.slice/apache2.service
              └─5385 /usr/sbin/apache2 -k start -DFOREGROUND
                └─5401 /usr/sbin/apache2 -k start -DFOREGROUND
                  └─5402 /usr/sbin/apache2 -k start -DFOREGROUND

Jun 29 04:16:08 Ubuntu systemd[1]: Starting apache2.service - The Apache HTTP Server...
Jun 29 04:16:08 Ubuntu apachectl[5385]: AH00558: apache2: Could not reliably determine the server's fully qualified domain name, please see the documentation on how to set 'ServerName'
Jun 29 04:16:08 Ubuntu systemd[1]: Started apache2.service - The Apache HTTP Server.
lines 1-18/18 (END)
```

27. Configure and secure a MySQL Database.

Installed, configured and secured a Mysql Database server

```
vboxuser@Ubuntu:~/Desktop$ sudo apt update && sudo apt install mysql-server -y
[sudo: authenticate] Password:
Hit:1 http://in.archive.ubuntu.com/ubuntu resolute InRelease
Hit:2 http://in.archive.ubuntu.com/ubuntu resolute-updates InRelease
Hit:3 http://in.archive.ubuntu.com/ubuntu resolute-backports InRelease
Hit:4 http://security.ubuntu.com/ubuntu resolute-security InRelease
Ign:5 https://pkg.jenkins.io/debian-stable binary/ InRelease
Hit:6 https://pkg.jenkins.io/debian-stable binary/ Release
6 packages can be upgraded. Run 'apt list --upgradable' to see them.
mysql-server is already the newest version (8.4.10-0ubuntu0.26.04.1).
Summary:
  Upgrading: 0, Installing: 0, Removing: 0, Not Upgrading: 6
vboxuser@Ubuntu:~/Desktop$ sudo systemctl status mysql
● mysql.service - MySQL Community Server
   Loaded: loaded (/usr/lib/systemd/system/mysql.service; enabled; preset: enabled)
   Active: active (running) since Mon 2026-06-29 04:38:00 UTC; 57min ago
     Invocation: 1b35568121414ce8ba174e82f5f7f66f
       Process: 1365 ExecStartPre=/usr/share/mysql/mysql-systemd-start pre (code=exited, status=0/SUCCESS)
    Main PID: 1649 (mysqld)
      Status: "Server is operational"
        Tasks: 34 (limit: 6371)
       Memory: 553.9M (peak: 554.4M)
         CPU: 55.865s
      CGroup: /system.slice/mysql.service
              └─1649 /usr/sbin/mysqld

Jun 29 04:37:46 Ubuntu systemd[1]: Starting mysql.service - MySQL Community Server...
Jun 29 04:38:00 Ubuntu systemd[1]: Started mysql.service - MySQL Community Server.
```

```
vboxuser@Ubuntu:~$ sudo mysql
[sudo: authenticate] Password:
Welcome to the MySQL monitor.  Commands end with ; or \g.
Your MySQL connection id is 8
Server version: 8.4.10-0ubuntu0.26.04.1 (Ubuntu)

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affiliates. Other names may be trademarks of their respective
owners.

Type 'help;' or '\h' for help. Type '\c' to clear the current input statement.

mysql> █
```

28. Set up a Application Server (e.g.,Apache Tomcat)

Installed prerequisite java and setup Apache tomcat application server.

```
vboxuser@Ubuntu:~$ sudo mysql
[sudo: authenticate] Password:
Welcome to the MySQL monitor.  Commands end with ; or \g.
Your MySQL connection id is 8
Server version: 8.4.10-0ubuntu0.26.04.1 (Ubuntu)

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affiliates. Other names may be trademarks of their respective
owners.

Type 'help;' or '\h' for help. Type '\c' to clear the current input statement.

mysql> █
```

```
vboxuser@Ubuntu:~/Desktop$ sudo apt install openjdk-17-jdk -y
Installing:
  openjdk-17-jdk

Installing dependencies:
  openjdk-17-jdk-headless  openjdk-17-jre  openjdk-17-jre-headless

Suggested packages:
  openjdk-17-demo  openjdk-17-source  visualvm  fonts-ipafont-gothic  fonts-ipafont-mincho  fonts-wqy-microhei |

Summary:
  Upgrading: 0, Installing: 4, Removing: 0, Not Upgrading: 6
  Download size: 123 MB
  Space needed: 276 MB / 16.3 GB available

Get:1 http://in.archive.ubuntu.com/ubuntu resolute-updates/universe amd64 openjdk-17-jre-headless amd64 17.0.19+10-
Get:2 http://in.archive.ubuntu.com/ubuntu resolute-updates/universe amd64 openjdk-17-jre amd64 17.0.19+10-1-26.04.2
Get:3 http://in.archive.ubuntu.com/ubuntu resolute-updates/universe amd64 openjdk-17-jdk-headless amd64 17.0.19+10-
Get:4 http://in.archive.ubuntu.com/ubuntu resolute-updates/universe amd64 openjdk-17-jdk amd64 17.0.19+10-1-26.04.2
Fetched 123 MB in 32s (3,860 kB/s)
Selecting previously unselected package openjdk-17-jre-headless:amd64.
```

```
Downloaded: 1 files, 14M in 1.7s (7.86 MB/s)
vboxuser@Ubuntu:~/Desktop$ sudo tar -xzf apache-tomcat-10.1.56.tar.gz
vboxuser@Ubuntu:~/Desktop$ sudo mv apache-tomcat-10.1.56 /opt/tomcat
vboxuser@Ubuntu:~/Desktop$ sudo chmod +x /opt/tomcat/bin/*.sh
chmod: cannot access '/opt/tomcat/bin/*.sh': No such file or directory
vboxuser@Ubuntu:~/Desktop$ cd /opt
vboxuser@Ubuntu:/opt$ sudo chmod +x /opt/tomcat/bin/*.sh
chmod: cannot access '/opt/tomcat/bin/*.sh': No such file or directory
vboxuser@Ubuntu:/opt$ /opt/tomcat/bin/startup.sh
bash: /opt/tomcat/bin/startup.sh: Permission denied
vboxuser@Ubuntu:/opt$ sudo /opt/tomcat/bin/startup.sh
Using CATALINA_BASE:   /opt/tomcat
Using CATALINA_HOME:   /opt/tomcat
Using CATALINA_TMPDIR: /opt/tomcat/temp
Using JRE_HOME:        /usr
Using CLASSPATH:        /opt/tomcat/bin/bootstrap.jar:/opt/tomcat/bin/tomcat-juli.jar
Using CATALINA_OPTS:
Tomcat started.
vboxuser@Ubuntu:/opt$ ps -ef | grep tomcat
root      12097      8958  29 06:00 ?        00:00:11 /usr/bin/java -Djava.util.logging.config.file=/opt/tomcat/conf/
ing.manager=org.apache.juli.ClassLoaderLogManager -Djdk.tls.ephemeralDHKeySize=2048 -Djava.protocol.handler.pkgs=or
n.io.useCanonicalCaches=false -Dorg.apache.catalina.security.SecurityListener.UMASK=0027 --add-opens=java.base/java.lan
/java.lang.reflect=ALL-UNNAMED --add-opens=java.base/java.io=ALL-UNNAMED --add-opens=java.base/java.util=ALL-UNNAME
oncurrent=ALL-UNNAMED --add-opens=java.rmi/sun.rmi.transport=ALL-UNNAMED -classpath /opt/tomcat/bin/bootstrap.jar:/
talina.base=/opt/tomcat -Dcatalina.home=/opt/tomcat -Djava.io.tmpdir=/opt/tomcat/temp org.apache.catalina.startup.B
vboxuser  12142      10466  0 06:01 pts/1    00:00:00 grep --color=auto tomcat
```

Home Documentation Configuration Examples Wiki Mailing Lists Find Help

Apache Tomcat/10.1.56

If you're seeing this, you've successfully installed Tomcat. Congratulations!



Recommended Reading:

- [Security Considerations How-To](#)
- [Manager Application How-To](#)
- [Clustering/Session Replication How-To](#)

[Server Status](#) [Manager App](#) [Host Manager](#)

Developer Quick Start

- [Tomcat Setup](#)
- [First Web Application](#)
- [Realms & AAA](#)
- [JDBC DataSources](#)
- [Examples](#)
- [Servlet Specifications](#)
- [Tomcat Versions](#)

Managing Tomcat

For security, access to the [manager webapp](#) is restricted. Users are defined in:

\$CATALINA_HOME/conf/tomcat-users.xml

In Tomcat 10.1 access to the manager application is split between different users.

Documentation

[Tomcat 10.1 Documentation](#)

[Tomcat 10.1 Configuration](#)

[Tomcat Wiki](#)

Find additional important configuration

Getting Help

[FAQ and Mailing Lists](#)

The following mailing lists are available:

[tomcat-announce](#)
Important announcements, releases, security vulnerability notifications. (Low volume).

29. Create a service file for Apache Tomcat.(Should execute by using systemctl command)

Created a service file for Apache tomcat by using systemctl commands.

```
vboxuser@Ubuntu:~$ sudo nano /etc/systemd/system/tomcat.service
vboxuser@Ubuntu:~$ sudo systemctl daemon-reload
vboxuser@Ubuntu:~$ sudo systemctl enable tomcat
Created symlink '/etc/systemd/system/multi-user.target.wants/tomcat.service' → '/etc/systemd/s
vboxuser@Ubuntu:~$ sudo systemctl start tomcat
Job for tomcat.service failed because the control process exited with error code.
See "systemctl status tomcat.service" and "journalctl -xeu tomcat.service" for details.
vboxuser@Ubuntu:~$ sudo systemctl start tomcat.service
Job for tomcat.service failed because the control process exited with error code.
See "systemctl status tomcat.service" and "journalctl -xeu tomcat.service" for details.
vboxuser@Ubuntu:~$ sudo systemctl status tomcat.service
● tomcat.service - Apache Tomcat Web Application Container
   Loaded: loaded (/etc/systemd/system/tomcat.service; enabled; preset: enabled)
   Active: activating (auto-restart) (Result: exit-code) since Mon 2026-06-29 10:46:04 UTC;
   Invocation: e935ad6b322b4dc2995e6a7b9b259dcc
     Process: 7554 ExecStart=/opt/tomcat/bin/startup.sh (code=exited, status=217/USER)
    Mem peak: 1.7M
       CPU: 15ms
```

30. Print specific columns from a delimited file.

To display only selected columns from a file where fields are separated by a delimiter (such as comma, colon, or tab).


```
vboxuser@Ubuntu:/opt$ cd
vboxuser@Ubuntu:~$ cut -d',' -f1,3 data.csv
cut: data.csv: No such file or directory
vboxuser@Ubuntu:~$ cut -d',' -f1,3 a.csv
cut: a.csv: No such file or directory
vboxuser@Ubuntu:~$ cut -d',' -f1,3 /a
cut: /a: No such file or directory
vboxuser@Ubuntu:~$ echo "101,Salman,Techie1
> 102,Sudeep,Techie2,
> 103,Sarfaraz,Techie3" > Trainees
vboxuser@Ubuntu:~$ ls
Desktop Documents Downloads Music Pictures Public Templates Trainees Videos home snap
vboxuser@Ubuntu:~$ cut -d',' -f1,3 Trainees
101,Techie1
102,Techie2
103,Techie3
vboxuser@Ubuntu:~$ cut -d',' -f1,2 Trainees
101,Salman
102,Sudeep
103,Sarfaraz
vboxuser@Ubuntu:~$
```

31. Filter and print lines based on a specific pattern or condition.

To quickly find required information from text files, logs, or configuration files using the grep command based on a specific pattern or condition.

```
vboxuser@Ubuntu:~$ echo "Raj,HR
> Salman,Techie
> Gisele,Admin" > Employees
vboxuser@Ubuntu:~$ echo "Raj,HR
Salman,Techie
Gisele,Admin" > Employees.txt
vboxuser@Ubuntu:~$ ls
Desktop Documents Downloads Employees Empl
vboxuser@Ubuntu:~$ cat Employees.txt
Raj,HR
Salman,Techie
Gisele,Admin
vboxuser@Ubuntu:~$ grep "Techie" Employees
Salman,Techie
vboxuser@Ubuntu:~$
```

32. Calculate and print the average, sum, or other statistics of a column.

To calculate the sum, average, count, minimum, or maximum values of a numeric column in a file.

```
vboxuser@Ubuntu:~$ echo " Salman 80
Saad 90
Saeed 100" > Marks1.txt
vboxuser@Ubuntu:~$ sudo awk '{sum += $2} END {print"Sum =", sum}' Marks1.txt
Sum = 270
vboxuser@Ubuntu:~$ sudo awk '{sum += $2} END {print"Average =", sum/NR}' Marks1.txt
Average = 90
vboxuser@Ubuntu:~$ awk END {print"Count =", NR}' Marks1.txt
> ^C
vboxuser@Ubuntu:~$ sudo awk END {print "Count =", NR}' Marks1.txt
> ^C
vboxuser@Ubuntu:~$ sudo awk 'END {print "Count =", NR}' Marks1.txt
Count = 3
vboxuser@Ubuntu:~$
```


33. Perform string manipulation, such as extracting substrings or changing case.

To manipulate strings in Linux by extracting substrings and converting text to uppercase or lowercase using commands like cut, awk, tr, and sed.

```
vboxuser@Ubuntu:~$ echo "Salman"
> Saud
> Saeed
> Saif" > Names.txt
vboxuser@Ubuntu:~$ cat Names.txt
Salman
Saud
Saeed
Saif
vboxuser@Ubuntu:~$ cut -c1-3 Names.txt
Sal
Sau
Sae
Sai
vboxuser@Ubuntu:~$ echo "linux administration" | tr 'a-z' 'A-Z'
LINUX ADMINISTRATION
vboxuser@Ubuntu:~$ echo "Salman" | awk '{print substr($1,4,5)}'
man
vboxuser@Ubuntu:~$
```

34. Count the occurrences of a specific pattern in a file.

The grep -c is used to count how many lines in a file contain a specific pattern. It is useful for quickly finding the number of matching entries in log files, configuration files, or text documents.

```
vboxuser@Ubuntu:~$ vi Employees
vboxuser@Ubuntu:~$ cat Employees
Raj,HR
Salman,Techie
Saif, Techie
Saud, Manager

vboxuser@Ubuntu:~$ grep -c "Techie" Employees
2
vboxuser@Ubuntu:~$ grep -c "Manager" Employees
1
```

35. Sort lines based on a specific field or column.

The sort command is used to arrange lines in a file. The -t option specifies the field delimiter, and the -k option specifies the field (column) to sort by

```
vboxuser@Ubuntu:~/Desktop$ echo "101,Saud,Manager
> 104,Salman,Techie
> 102,Saeed,Admin
> 103,Saif,Techie" > Employees
vboxuser@Ubuntu:~/Desktop$ sort -t ',' -k1 Employees
101,Saud,Manager
102,Saeed,Admin
103,Saif,Techie
104,Salman,Techie
vboxuser@Ubuntu:~/Desktop$
```

36. Merge multiple files based on a common field or column.

The join command merges two sorted files by comparing a common field (column).

Both files must be sorted on the join field before using the join command.

```
vboxuser@Ubuntu:~$ echo "101 Raj
> 102 Simran
> 103 Sanjay
> 104 Anil" > Jobs
vboxuser@Ubuntu:~$ echo "101 HR
> 102 IT
> 103 Sales
> 104 Admin" > Department
vboxuser@Ubuntu:~$ cat Jobs Department
101 Raj
102 Simran
103 Sanjay
104 Anil
101 HR
102 IT
103 Sales
104 Admin
vboxuser@Ubuntu:~$ sort Jobs -o Jobs
vboxuser@Ubuntu:~$ sort Department -o Department
vboxuser@Ubuntu:~$ join -1 1 -2 1 Jobs Department
101 Raj HR
102 Simran IT
103 Sanjay Sales
104 Anil Admin
vboxuser@Ubuntu:~$
```

37. Substitute text in a file using search and replace.

The sed (Stream Editor) command is used to search for a specific word or pattern and replace it with another word. It can display the modified output or update the file directly using the -i option.

```

vboxuser@Ubuntu:~$ ls
Department Documents Employees Jobs Marks.txt Music
Desktop Downloads Employees.txt Marks Marks1.txt Names.txt
vboxuser@Ubuntu:~$ cat Employees
Raj,HR
Salman,Techie
Saif, Techie
Saud, Manager

vboxuser@Ubuntu:~$ sudo sed 's/Manager/Teamlead/' employ
[sudo: authenticate] Password:
sed: can't read employ: No such file or directory
vboxuser@Ubuntu:~$ sudo sed 's/Manager/Teamlead/' Employees
Raj,HR
Salman,Techie
Saif, Techie
Saud, Teamlead

vboxuser@Ubuntu:~$ sudo sed -i 's/Manager/Teamlead/g' Employees
vboxuser@Ubuntu:~$ cat Employees
Raj,HR
Salman,Techie
Saif, Techie
Saud, Teamlead

```

38. Delete specific lines based on a pattern or line number.

The sed command is used to delete lines from a file. You can delete a line by specifying its line number or by matching a specific pattern.

```

vboxuser@Ubuntu:~$ cat Employees
Raj,HR
Salman,Techie
Saif, Techie
Saud, Teamlead

vboxuser@Ubuntu:~$ sed '3d' Employees
Raj,HR
Salman,Techie
Saud, Teamlead

vboxuser@Ubuntu:~$ sed -i '3d' Employees
vboxuser@Ubuntu:~$ cat Employees
Raj,HR
Salman,Techie
Saud, Teamlead

vboxuser@Ubuntu:~$ sed '/Teamlead/d' Employees
Raj,HR
Salman,Techie

vboxuser@Ubuntu:~$ sed -i '/Teamlead/d' Employees
vboxuser@Ubuntu:~$ cat Employees
Raj,HR
Salman,Techie

```

39. Append or insert text before or after a specific pattern or line.

The sed command is used to insert or append text in a file. The i option inserts text before a line or pattern, while the option appends text after a line or pattern.

```
vboxuser@Ubuntu:~$ ls
Department Documents Employees Jobs Marks.t
Desktop Downloads Employees.txt Marks Marks1.
vboxuser@Ubuntu:~$ cat Names.txt
Salman
Saud
Saeed
Saif
vboxuser@Ubuntu:~$ sed 'i\Sameer' Names.txt
Sameer
Salman
Saud
Saeed
Saif
vboxuser@Ubuntu:~$ sed '2a\Abd' Names.txt
Salman
Saud
Abd
Saeed
Saif
vboxuser@Ubuntu:~$ sed -i '2a\Abd' Names.txt
vboxuser@Ubuntu:~$ cat Names.txt
Salman
Saud
Abd
Saeed
Saif
vboxuser@Ubuntu:~$
```

40. Print only specific lines from a file.

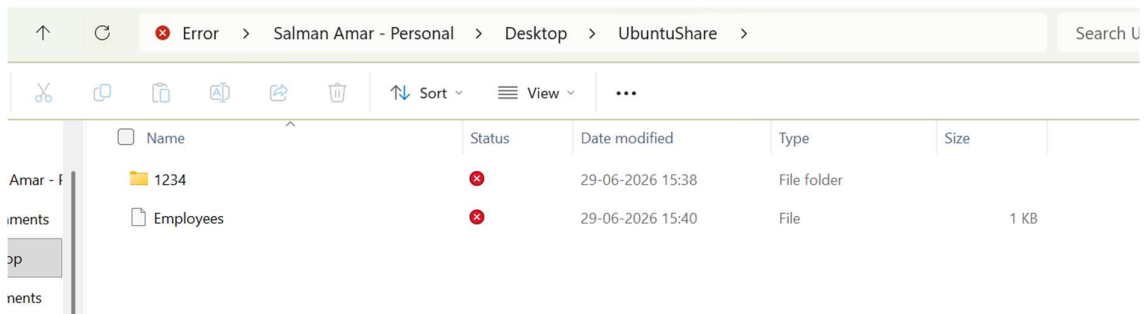
The sed command is commonly used to print specific lines from a file. You can specify a single line number or a range of line numbers.

```
vboxuser@Ubuntu:~$ ls
Department Documents Employees Jobs Ma
Desktop Downloads Employees.txt Marks Ma
vboxuser@Ubuntu:~$ cat Names.txt
Salman
Saud
Abd
Saeed
Saif
vboxuser@Ubuntu:~$ sed -n '3p' Names.txt
Abd
vboxuser@Ubuntu:~$ sed -n '2,4p' Names.txt
> ^C
vboxuser@Ubuntu:~$ sed -n '2,4p' Names.txt
Saud
Abd
Saeed
vboxuser@Ubuntu:~$ sed -n '$p' Names.txt
Saif
vboxuser@Ubuntu:~$
```


41. Copy file from linux to windows machine

To copy a file from an Ubuntu virtual machine to a Windows machine
using VMware Shared Folders

```
vboxuser@Ubuntu:~$ lsmod | grep vbox
vboxguest                57344  0
vboxuser@Ubuntu:~$ mount | grep sf_
vboxuser@Ubuntu:~$ sudo mount | grep sf_
[sudo: authenticate] Password:
vboxuser@Ubuntu:~$ sudo mkdir -p /mnt/share && sudo mount -t vboxsf UbuntuShare /mnt/share
vboxuser@Ubuntu:~$ ls /mnt/share
vboxuser@Ubuntu:~$ ls /mnt/share
Employees
vboxuser@Ubuntu:~$ ls
Department  Documents  Employees  Jobs  Marks.txt  Music  Pictures  Templates  UbuntuShare  home
Desktop    Downloads  Employees.txt  Marks  Marks1.txt  Names.txt  Public  Trainees  Videos  snap
vboxuser@Ubuntu:~$ sudo cp ~/home/Employees /mnt/share
cp: cannot stat '/home/vboxuser/home/Employees': Not a directory
vboxuser@Ubuntu:~$ sudo cp ~/Employees /mnt/share
vboxuser@Ubuntu:~$ ls /mnt/share
Employees
vboxuser@Ubuntu:~$
```



42. 5 use cases for AWK and 5 use cases for sed

```
vboxuser@Ubuntu:~$ echo " Salman 80
Saud 90
Saeed 100" > Marks1.txt
vboxuser@Ubuntu:~$ sudo awk '{sum += $2} END {print"Sum =", sum}' Marks1.txt
Sum = 270
vboxuser@Ubuntu:~$ sudo awk '{sum += $2} END {print"Average =", sum/NR}' Marks1.txt
Average = 90
vboxuser@Ubuntu:~$ awk END {print"Count =", NR}' Marks1.txt
> ^C
vboxuser@Ubuntu:~$ sudo awk END {print "Count =", NR}' Marks1.txt
> ^C
vboxuser@Ubuntu:~$ sudo awk 'END {print "Count =", NR}' Marks1.txt
Count = 3
vboxuser@Ubuntu:~$
```

```
vboxuser@Ubuntu:~$ ls
Department  Documents  Employees  Jobs  Ma
Desktop     Downloads  Employees.txt  Marks  Ma
vboxuser@Ubuntu:~$ cat Names.txt
Salman
Saud
Abd
Saeed
Saif
vboxuser@Ubuntu:~$ sed -n '3p' Names.txt
Abd
vboxuser@Ubuntu:~$ sed -n 2,4p' Names.txt
> ^C
vboxuser@Ubuntu:~$ sed -n '2,4p' Names.txt
Saud
Abd
Saeed
vboxuser@Ubuntu:~$ sed -n '$p' Names.txt
Saif
vboxuser@Ubuntu:~$
```